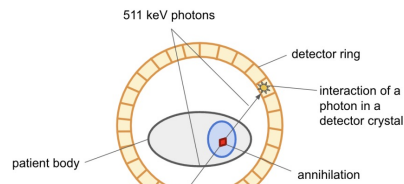


## Introduction

Positron Emission Tomography (PET) is an imaging technique that provides information about metabolic and functional processes in the body. The non-invasive technique works by injecting a radioactive tracer into the patient's blood stream. After having accumulated in specific regions of the body, the tracer undergoes radioactive decay and releases a positron. When the positron collides with an electron, they annihilate, releasing two gamma photons in nearly opposite directions. A PET scanner detects the nearly simultaneous arrival of the two photons with detectors arranged in a ring. With many recorded events, clever algorithms can be used to estimate points of annihilation and produce a three-dimensional image, a process that is called *image reconstruction*.

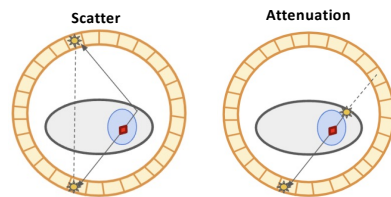


## Problem

Image reconstruction is complicated by two phenomena—scatter and attenuation. *Scatter* refers to a gamma photon changing direction after interacting with patient tissue, which leads to an incorrect localization of its origin. *Attenuation* refers to a gamma photon's absorption or loss of signal strength before reaching the detector, reducing the number of true coincident events. Both scatter and attenuation cause a reduction in quality of the reconstructed image.

Current image reconstruction methods correct for scatter and attenuation using attenuation maps and iterative algorithms, which are computationally expensive. A key insight is that information related to scatter and attenuation is already present in the raw PET data. Namely, photon energy measurements can help identify scatter events, and time-of-flight information improves localization along each line of response, reducing ambiguities caused by attenuation. This suggests that deep learning could potentially be leveraged to perform a more efficient and more accurate scatter and attenuation correction.

**Research Question:** Can a deep learning model be trained to perform scatter and attenuation correction without an attenuation map?

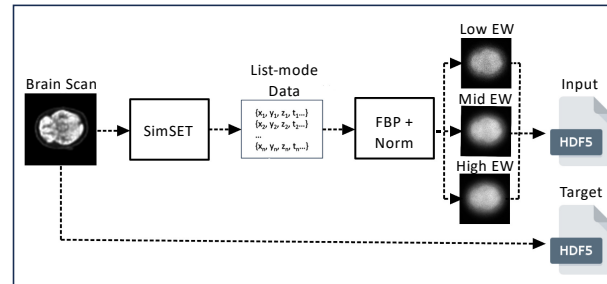


## Methods

The pipeline for processing the training data was as follows:

1. Patient brain scans were run through SimSET, a simulation tool, to generate list-mode data characteristic of the Gen2 system, a new brain imaging system.
2. The coincident photon detections were split into 3 energy groups: *low energy window* (both photons are below 511 keV), *middle energy window* (one photon is below 511 keV and the other is above 511 keV), and *high energy window* (both photons are above 511 keV).
3. The three energy windows were back-projected once to obtain three blurry images, and corresponding slices were saved as an input file.
4. The original ground truth slice was saved as a target file.

Training Data Processing Pipeline

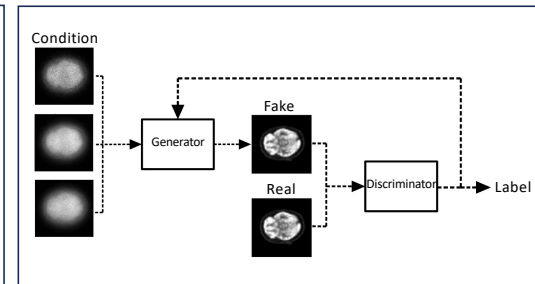


Using the processed input and target files, a Pix2Pix model, which is a conditional Generative Adversarial Network (cGAN), was trained to map the low, middle, and high energy window images to the scatter and attenuation corrected image. The learning architecture consists of a generator, which learns to produce fake data, and a discriminator, which learns to differentiate between real and fake data.

Training Data/Metrics:

- The training dataset was composed of 28 patient scans.
- The first and last 10 slices of each scan were removed.
- Training lasted for 200 epochs (~7 hours).

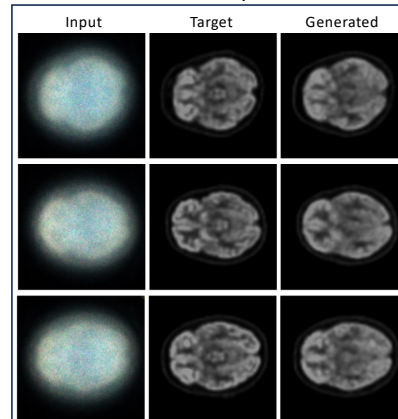
cGAN Architecture



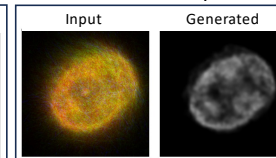
## Results

The trained model was tested using 9 out-of-sample simulated patient scans and 1 non-simulated patient scan. The simulated results were assessed quantitatively, using Peak Signal to Noise Ratio and Structural Similarity Index Measure. The non-simulated scan was not normalized.

Simulated Inputs



Non-Simulated Input



Quantitative Metrics

Average PSNR: 25.23 dB  
Average SSIM: 0.92

## Future Steps

1. Train a model with a larger and higher quality dataset.
2. Succeed in normalizing non-simulated data.
3. Clean up and extensively document the Deep Learning – Filtered Backprojection pipeline.
4. Develop a new pipeline solely for inferencing.

## References

- Dong, Q., Ullah, M. N., Innes, D., Watkins, R. D., Chang, C.-M., Zou, S. J., Groll, A., Sacco, I., Chinn, G., & Levin, C. S. (2024). PETcoil: First results from a second-generation RF-penetrable TOF-PET brain insert for simultaneous PET/MRI. *Physics in Medicine & Biology*, 69(18), 185007. <https://doi.org/10.1088/1361-6560/ad7221>
- Isola, P., Zhu, J.-Y., Zhou, T., & Efros, A. A. (2017). Image-to-image translation with conditional adversarial networks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 5967–5976. <https://doi.org/10.1109/CVPR.2017.632>